Manav Vora

Reinforcement Learning | Diffusion Models | Multi-Agent Systems

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Ph.D. researcher developing reinforcement learning and diffusion methods for scalable, intelligent multi-agent systems. Research spans communication learning, meta-RL, and generative motion planning, with industry experience in LLM post-training at Nokia Bell Labs and hardware-efficient motion prediction at Rivian.

EDUCATION

University of Illinois Urbana-Champaign (UIUC) – Ph.D., Aerospace Engineering (Robotics and AI)
Aug 2022 – Expected 2027 | GPA: 3.9/4.0 | Advisor: Prof. Melkior Ornik.

Indian Institute of Technology Bombay (IITB) – B.Tech. (Honors), Aerospace Engineering
Aug 2018 – Aug 2022 | GPA: 4.0/4.0 | Department rank: 3. Minor: Systems and Controls Engineering.

RESEARCH AND INDUSTRY EXPERIENCE

University of Illinois Urbana-Champaign
Research Assistant, LEADCAT Lab | Aug 2022 – Present

- Developed **Akule** [1], a compositional diffusion framework for scalable joint multi-robot motion planning with learned sparse interactions around a frozen single-robot prior; achieved **up to 25× end-to-end planning speedup**, zero-shot transfer across team sizes, and planning with 256 robots.
- Designed **SCoUT** [3] for learned communication in multi-agent reinforcement learning (MARL), using temporal grouping and counterfactual credit assignment; achieved **100% win rate** in competitive two-team MARL benchmarks with 100 agents/team.
- Developed **meta-RL and resource-allocation algorithms** [2, 5, 10] for budget-constrained sequential decisions, scaling to 1,000-component POMDPs and 1,000-robot MDPs.
- Developed **B-COD** [4], coupling belief-conditioned one-step diffusion with Soft Actor-Critic (SAC) for adaptive sensing; achieved 97.9% goal reach with **57.7% less sensing energy** than always-on sensing.
- Developed **ComTraQ-MPC** [9], coupling meta-trained RL with model predictive control to schedule budgeted sensing and track trajectories; validated on TurtleBot hardware and unseen trajectory–budget pairs.
- Developed **graph-based planning and optimization** for unknown nonlinear dynamics using online system identification and predictive reachability [7]; designed resource-efficient modular-agent routing [6] and environment layouts for earlier adversarial goal recognition [8].
- **Ongoing:** extending diffusion-based coordination to heterogeneous search and rescue; developing long-tail trajectory prediction under occlusion and multi-agent behavior inference from partial, top-down observations.

Nokia Bell Labs – Artificial Intelligence Lab
Machine Learning & AI Research Intern | Jun 2026 – Aug 2026

- Developed a **world-model-guided framework for multi-agent conflict resolution**, combining bounded LLM contracts, counterfactual evaluation, and log-Nash optimization under external hard safety constraints.
- Post-trained Qwen2.5 with LoRA and clipped GRPO using downstream decision rewards, achieving **89.6% normalized decision quality** versus 73.3% (16-shot direct-action ICL), 59.0% (SFT), and 47.1% (base).
- Built a 1,000-case data-generation, reference-contract search, and evaluation pipeline; reduced safety-recovery interventions from **17.3% to 4.0%** versus the same ICL baseline on 100 held-out simulated O-RAN cases.

Rivian
Machine Learning Intern | May 2025 – Aug 2025

- Extended **HPTR/Wayformer** for multimodal multi-agent forecasting; trained and benchmarked on Argoverse 2.
- Designed INT16 post-training quantization achieving **2× inference speedup over FP32**, with $<0.01\%$ mAP degradation and $<0.1\%$ minADE increase.
- Developed mixed INT8/INT16 quantization with automated layer-wise sensitivity calibration, reducing inference latency by $\approx$**12%** with $\approx 0.02\%$ accuracy loss.

Corteva Agriscience / UIUC
Industry-Sponsored Research | Aug 2025 – Dec 2025

- Developed **MAESTRO**, an open-source multi-agent system that turns supply-chain optimization outputs into transparent, source-grounded explanations for nontechnical stakeholders.
- Implemented **agentic retrieval-augmented generation (RAG)** in LangGraph with expert routing, iterative retrieval, tool use, and validation; aggregated specialized analyses into grounded responses.
- Integrated Pandas/Excel tools for programmatic KPI and throughput analysis of solver spreadsheets; delivered a stakeholder-facing report and evaluation artifacts.

SELECTED PUBLICATIONS AND PREPRINTS

[1] **M. Vora**, G. Puthumanaillam, A. Penumarti, J. Shin, and M. Ornik. **Akule: Fast and Scalable Multi-Robot Motion Planning via Sparse Interaction Diffusion**. *arXiv:2610.14248, 2026 (Under review at AAMAS 2027).*

[2] **M. Vora**, J. Liang, M. N. Grussing, and M. Ornik. **Solving Truly Massive Budgeted Monotonic POMDPs with Oracle-Guided Meta-Reinforcement Learning**. *Transactions on Machine Learning Research (TMLR), 2026.*

[3] **M. Vora**, G. Puthumanaillam, H. Tsukamoto, and M. Ornik. **SCoUT: Scalable Communication via Utility-Guided Temporal Grouping in Multi-Agent Reinforcement Learning**. *Reinforcement Learning Journal (RLJ) / Reinforcement Learning Conference (RLC), 2026.* ***Oral presentation***.

[4] G. Puthumanaillam, A. Penumarti, **M. Vora**, P. Padrao, J. Fuentes, L. Bobadilla, J. Shin, and M. Ornik. **Belief-Conditioned One-Step Diffusion: Real-Time Trajectory Planning with Just-Enough Sensing**. *Conference on Robot Learning (CoRL), 2025.* ***Oral presentation***.

[5] **M. Vora**, I. Shomorony, and M. Ornik. **Capacity-Aware Planning and Scheduling in Budget-Constrained Multi-Agent MDPs: A Meta-RL Approach**. *IEEE Robotics and Automation Letters (RA-L), 2025.*

[6] A. Casselman, **M. Vora**, and M. Ornik. **Virtual Force-Based Routing of Modular Agents on a Graph**. *IEEE Conference on Control Technology and Applications (CCTA), 2026.*

[7] Z. Zhang, G. Puthumanaillam, **M. Vora**, and M. Ornik. **Motion Planning and Control with Unknown Nonlinear Dynamics through Predicted Reachability**. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.* ***Oral presentation***.

[8] A. Dey, S. Ruggerio, **M. Vora**, and M. Ornik. **Optimizing a Model-Agnostic Measure of Graph Counterdeceptiveness via Reattachment**. *IEEE Cyber Awareness and Research Symposium (CARS), 2025.*

[9] G. Puthumanaillam*, **M. Vora***, and M. Ornik. **ComTraQ-MPC: Meta-Trained DQN-MPC Integration for Trajectory Tracking with Limited Active Localization Updates**. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.* ***Oral presentation***.

[10] **M. Vora**, P. Thangeda, M. N. Grussing, and M. Ornik. **Welfare Maximization Algorithm for Solving Budget-Constrained Multi-Component POMDPs**. *IEEE Control Systems Letters (LCSS), 2023.*

*Equal contribution.

TECHNICAL SKILLS

Languages / ML: Python, C++, Julia, MATLAB; PyTorch, TensorFlow, Stable-Baselines3, AIMET.
Robotics / tools: ROS/ROS2, OpenCV, Gazebo, LangGraph, MLflow, Git, ChromaDB, Gym

HONORS AND AWARDS

Cozad New Venture Challenge (2026): UncertIA team, third place; $125K in prizes and investment.
UIUC Aerospace Engineering Department Fellowship (2023), for research and academic achievements.
IEEE CDC Student Travel Award (2023); **Chanakya Research Fellowship (2021)**.

TEACHING, MENTORSHIP AND SERVICE

Teaching Assistant, UIUC: AE 498 (Fall 2026); AE 353 (Spring 2026); AE 352 (Spring 2024, 2025).
Reviewer: ICLR, L4DC, IEEE RA-L, ICRA, IROS, AAMAS, CoRL
Mentoring four undergraduate students and 2 graduate students.